

Computational Intelligence for Optimization

Master in Data Science and Advanced Analytics

Approximating Vermeer's *Girl with a Pearl Earring* with 100 Triangles

A Genetic Algorithm Approach



Best result achieved: RMSE = 25.96 (100 triangles, 5 000 generations)

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1. Introduction

The aim of this project is to produce a recognisable, minimalist triangle-based interpretation of Vermeer's Girl with a Pearl Earring. A canvas of 300×400 pixels is composed exclusively of 100 opaque, single-colour triangles whose shapes, positions and colours are evolved by a Genetic Algorithm (GA). Visual resemblance is the ultimate objective; pixel-wise Root Mean Squared Error (RMSE) serves as the computational proxy that guides the search, lower RMSE consistently corresponds to a more visually faithful rendering.

This report describes the representation chosen for the triangles, the fitness function, and the operator design process. The latter is grounded in a systematic ablation study (Challenge 3) comprising 300 independent runs across all operator combinations, enabling data-driven justification of every design decision.

2. Representation

2.1 Triangle Encoding

Each triangle is encoded as a vector of nine continuous floats: $[x_1, y_1, x_2, y_2, x_3, y_3, R, G, B]$. Vertex coordinates are sampled uniformly from $[0, 300]$ (x) and $[0, 400]$ (y); colour channels from $[0, 255]$. An individual is a list of 100 such vectors, a genome of 900 real-valued genes. Triangles are rendered in sequence onto a black canvas using PIL's polygon routine; later triangles paint over earlier ones (painter's algorithm). All genes remain continuous throughout evolution.

2.2 Representation Variants Compared

Three initialisation and encoding strategies were evaluated and compared visually as well as quantitatively after 800 generations (10 runs each, pop = 80):

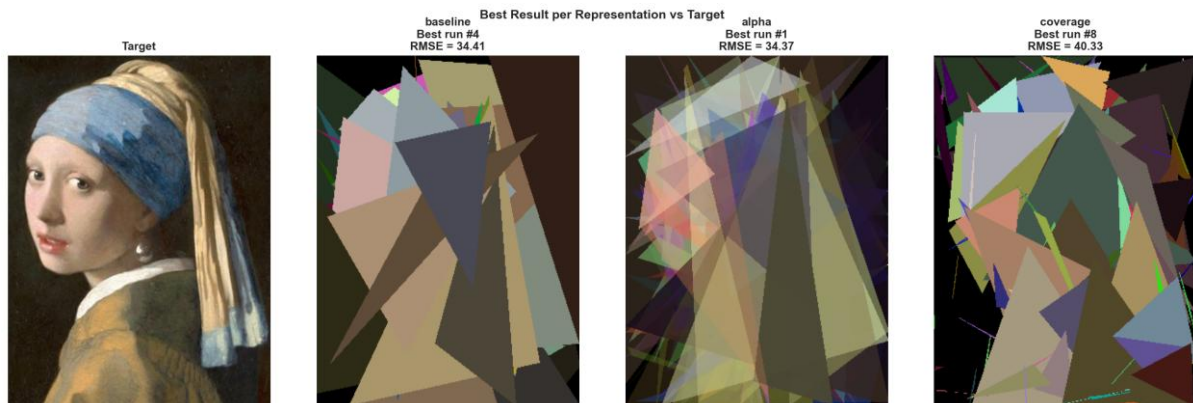


Figure 1 – Best visual result per representation variant after 800 generations. Left to right: original painting; Baseline RGB (RMSE = 34.41); Alpha RGBA (RMSE = 34.37); Coverage init (RMSE = 40.33).

- **Baseline (RGB, 9 genes per triangle):** Standard unconstrained initialisation. After 800 generations it already captures the dominant colour masses and rough facial outline of the painting, achieving RMSE = 34.41 in the best run.
- **Alpha variant (RGBA, 10 genes per triangle):** An additional alpha channel (initialised in $[50, 220]$) enables semi-transparent blending between triangles. This richer representation achieves RMSE = 34.37 at 800 generations, slightly better with lower inter-run variance ($\sigma = 0.96$ vs 1.82). However, the RGBA renderer composites a separate layer for each of the 100 triangles

per fitness evaluation, making each call substantially more expensive than the standard RGB renderer. Combined with the absence of fitness caching, this rendered 5 000-generation runs and the 30 independent trials required for reliable statistical testing computationally prohibitive; the variant was therefore retained as an exploratory comparison only.

- **Coverage initialisation (RGB, 9 genes):** Triangles were assigned to cells of a 10×10 grid, with vertices constrained within each 30×40 px cell. Despite the intent of guaranteeing full spatial coverage from generation 0, the resulting small triangles cannot approximate large colour regions, leading to a higher initial RMSE (~108 vs ~97 for baseline) and consistently worse final performance (RMSE = 40.33 best). As visible in Figure 1, the image is noticeably more fragmented and less recognisable. This strategy was conclusively ruled out.

A centroid + angle + scale parameterisation was also considered but rejected: it offers no expressive advantage over direct vertex coordinates and introduces trigonometric non-linearities that complicate mutation without benefit.

3. Fitness Function

Fitness is the pixel-wise RMSE between the rendered image and the target, computed across all $300 \times 400 \times 3 = 360\,000$ channel values in the range $[0, 255]$:

$$\text{RMSE} = \sqrt{\left[\frac{1}{N} \cdot \sum (\text{rendered_i} - \text{target_i})^2 \right]}$$

The objective is minimisation. RMSE is a natural proxy for visual resemblance: reducing it consistently pushes the triangle composition towards the colour and luminance structure of the original painting, as confirmed by the visual progression observable across generations. RMSE values are cached per individual to avoid redundant renders during selection; the target image array is loaded once at module level. Typical initial RMSE is ~97; the best final value achieved is 25.96.

4. GA Operator Design and Ablation Study

To justify operator choices rigorously, all 30 combinations (2 mutation × 5 selection × 3 crossover variants) were run 10 times each (300 runs total, pop = 80, 800 generations, xo_prob = 0.7, mut_prob = 0.05, elitism = 2). Each component was then evaluated by marginalising over the other two, averaging for a given variant, all runs that used it regardless of the remaining operators.

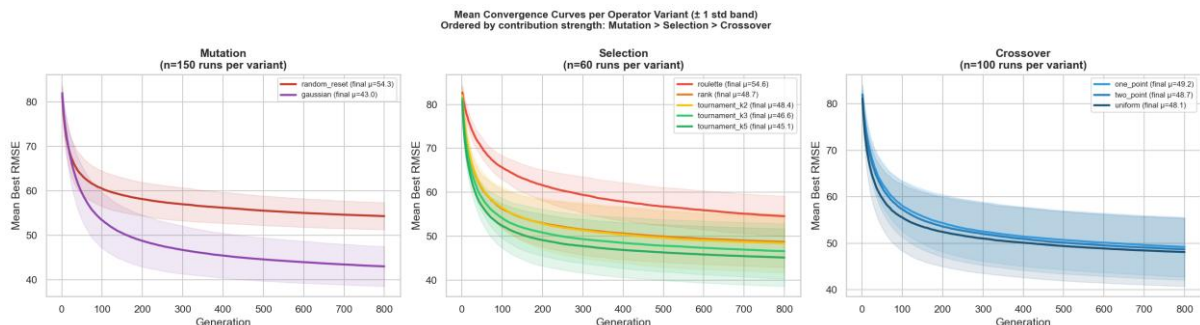


Figure 2 – Marginal mean convergence curves (± 1 std band) per operator variant. Left: Mutation (dominant factor). Centre: Selection (moderate effect). Right: Crossover (minimal effect).

4.1 Mutation — Dominant Factor

Mutation type is by far the most influential component. Gaussian mutation, where each gene is independently perturbed with probability mut_prob by noise $N(0, \sigma = 15)$, followed by clamping to gene bounds, consistently and decisively outperforms random-reset mutation throughout all 800 generations, across all selection and crossover combinations (Figure 2, left panel). Random-reset replaces entire triangles with freshly randomised ones, destroying fine-grained spatial and colour refinements accumulated by prior generations. Gaussian perturbation enables incremental local search, which is essential for a 900-dimensional continuous landscape. Gaussian mutation was adopted exclusively.

4.2 Selection — Moderate Effect

Tournament selection consistently outperforms both fitness-proportionate (roulette) and rank-based selection. The improvement is monotonic with tournament size k : $k = 5$ yields the best marginal mean. Roulette wheel selection, even with inverse-fitness weighting for minimisation, loses discriminative power as the population converges and fitness values cluster, whereas tournament maintains constant selection pressure regardless of absolute fitness spread. Tournament $k = 5$ was selected.

4.3 Crossover — Minimal Impact

All three crossover operators act at the triangle level, swapping whole triangle vectors between parents rather than splitting within a triangle's nine genes. This design choice preserves triangle coherence: mixing the vertex positions of one triangle with the colour of another produces geometrically inconsistent offspring. One-point, two-point and uniform crossover produce nearly identical marginal means with heavily overlapping confidence bands (Figure 2, right panel). All three appear in the top-5 configurations of the grid search. Uniform crossover was retained for marginal superiority.

5. Statistical Validation and Probability Tuning

5.1 Statistical validation of operator choice

The top three operator configurations from the grid search were re-run for 30 independent trials each, the minimum required for reliable non-parametric testing. Population size was raised to 100 (from 80 in the grid search phase): with the operator space narrowed to three candidates and a clearer optimisation direction, the larger population reduces noise and provides a more stable baseline for the subsequent probability tuning. A Kruskal-Wallis test confirmed an overall significant difference among the three configurations ($p < 0.05$), justifying pairwise comparisons. Mann-Whitney U tests with Bonferroni correction (adjusted threshold $\alpha = 0.0167$) showed that *tournament $k = 5$ | uniform | gaussian* (mean RMSE 37.54 ± 2.05) is significantly better than *tournament $k = 5$ | two-point | gaussian* (38.96 ± 1.92 , $p = 0.0108$); the remaining pairwise comparisons did not reach significance. The winning configuration was carried forward as the fixed baseline for probability tuning.

5.2 Probability tuning

The Schema Theorem (Holland, 1975) provides the theoretical basis for choosing crossover and mutation probabilities. Each schema H survives mutation with probability $P_s \approx 1 - o(H) \cdot p_m$, where $o(H)$ is the schema order. The Building Blocks Hypothesis extends this: high-fitness, low-order, short-length schemas grow exponentially in the population and are recombined by crossover into progressively better solutions. Two crossover probabilities $\{0.7, 0.9\}$ and three mutation rates $\{0.01, 0.05, 0.10\}$ were tested, giving the schema-survival estimates shown in Table 1.

xo_prob	mut_prob	Schema survival (o = 5)	Mean RMSE	Std	Best RMSE
0.9 ★	0.01 ★	0.95	33.06	2.60	29.94
0.7	0.01	0.95	33.96	1.85	30.79
0.9	0.05	0.75	36.41	1.57	33.57
0.7	0.05	0.75	37.40	2.64	34.05
0.9	0.10	0.50	40.26	1.36	37.66
0.7	0.10	0.50	40.50	1.27	38.41

Table 1 – Probability tuning results (10 runs, 800 generations, pop = 100). ★ = chosen configuration. Schema survival computed as $1 - o(H) \cdot p_m$ for reference order $o = 5$.

Mutation rate is the dominant factor: mean RMSE degrades monotonically as p_m rises from 0.01 to 0.10 at both crossover levels, consistent with the schema-preservation prediction. At $p_m = 0.10$, half of all order-5 schemas are destroyed per generation, effectively reducing the search to near-random perturbation in this 900-dimensional space. Crossover probability has a secondary but consistent effect: xo_prob = 0.9 outperforms 0.7 at every mutation level, confirming that more frequent recombination amplifies the building-block combination mechanism. The configuration **xo_prob = 0.9, mut_prob = 0.01** was selected.

5.3 Final Configuration Summary

Component	Chosen Setting	Alternatives Tested	Effect
Selection	Tournament k = 5	Roulette, Rank, k ∈ {2,3}	Moderate
Crossover	Uniform (triangle-level)	One-point, Two-point	Low
Mutation	Gaussian σ = 15	Random-reset	Dominant
Elitism	2 individuals	—	—
xo_prob	0.90	0.70	Moderate
mut_prob	0.01	0.05, 0.10	High

Table 2 – Final GA configuration and ablation summary.

6. Island Model

An island model GA was implemented as an architectural enhancement. Four sub-populations of 25 individuals evolve independently (total budget: 100 individuals, equal to the standard GA). A ring migration topology transfers the best individual from each island to its clockwise neighbour every 20 generations. Operators and hyperparameters were identical to the final standard GA configuration (tournament k = 5, uniform crossover, Gaussian mutation, xo = 0.9, mut = 0.01, elitism 2 per island); 10 runs at 5 000 generations were performed.

7. Final Results

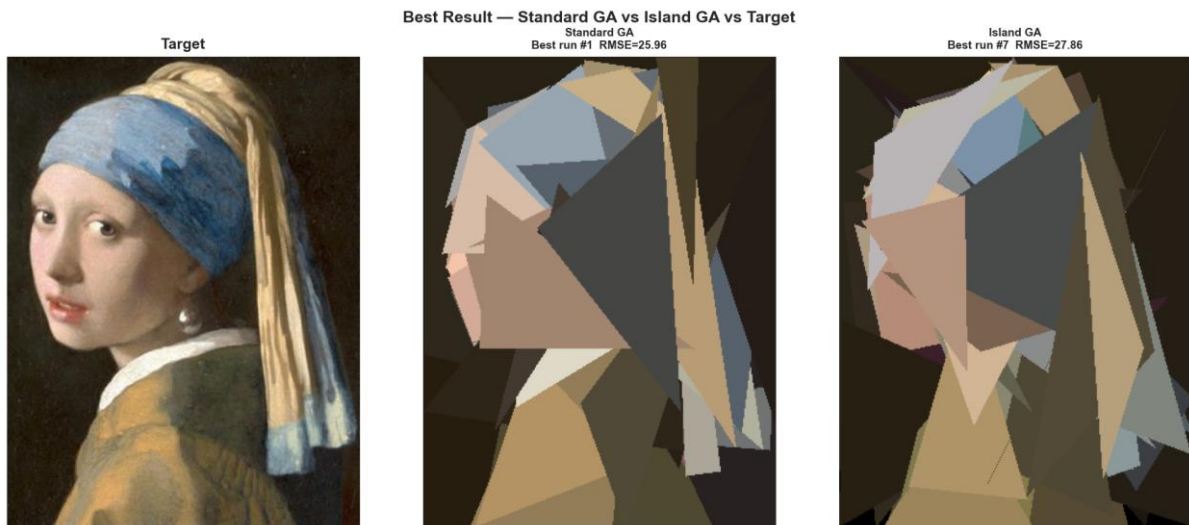


Figure 3 – Visual comparison: original painting (left), best Standard GA result run #1, RMSE = 25.96 (centre), best Island GA result run #7, RMSE = 27.86 (right). All from 10 independent runs of 5 000 generations.

Figure 3 illustrates the main outcome of the project: with 100 triangles and 5 000 generations, the GA produces a composition in which the subject of the painting is clearly identifiable, the dark background, the profile of the face, the blue-and-white headscarf, the pearl earring area, and the illuminated collar are all discernible. The Standard GA produces the sharper and more faithful result; the Island GA, while slightly less detailed, is also visually recognisable.

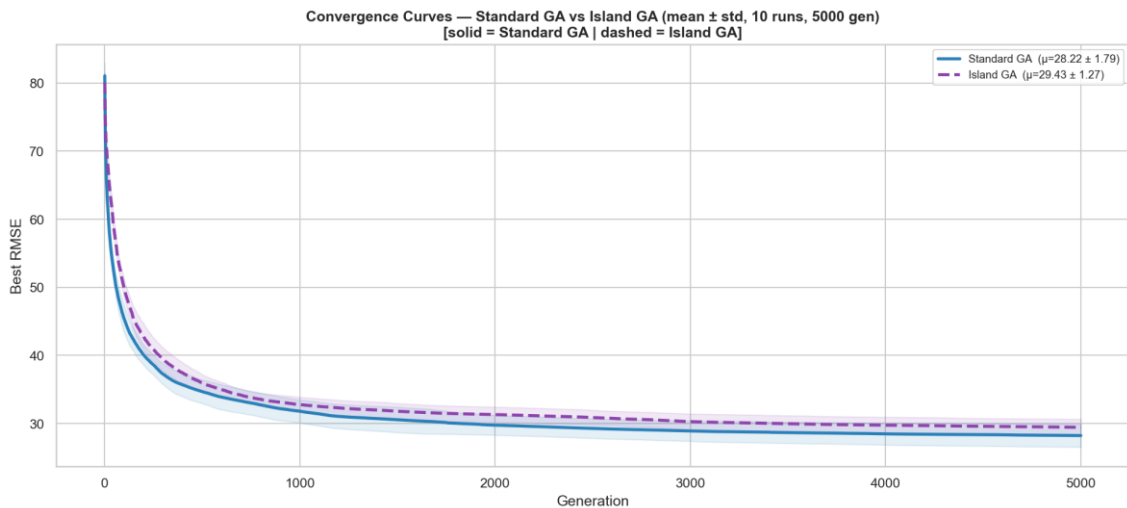


Figure 4 – Convergence curves (mean ± std, 10 runs). Standard GA (solid blue) reaches lower final RMSE than the Island GA (dashed purple), though the Island GA shows tighter variance.

Configuration	Pop.	Mean RMSE	Std	Best RMSE	Worst RMSE
Standard GA	100	28.22	1.70	25.96	32.26
Island GA (4 × 25)	100	29.43	1.20	27.86	31.63

Table 3 – Standard GA vs Island GA (10 runs, 5 000 generations).

The Standard GA achieves the lower mean and best RMSE. The Island GA is more consistent (lower standard deviation and better worst case), suggesting a robustness advantage, but the sub-populations of 25 individuals with very low mutation (0.01) converge too quickly intra-island for migration to compensate. The standard single-population GA sustains diversity across 100 individuals for longer, enabling greater refinement.

Run	Initial	@1 000	@2 000	@3 000	@4 000	Final RMSE
01	96.01	33.35	30.75	28.07	26.46	25.96 ★
02	98.22	32.44	30.54	29.30	28.53	28.29
03	96.93	32.66	30.10	29.38	29.01	28.84
04	94.57	30.40	29.01	28.01	27.70	27.05
05	98.60	30.19	28.77	28.20	28.00	27.88
06	97.25	30.65	29.01	28.51	28.32	28.27
07	96.70	35.51	33.06	32.60	32.38	32.26
08	100.49	29.52	27.80	27.00	26.86	26.74
09	96.45	31.24	28.11	27.73	27.43	27.17
10	100.27	31.99	30.30	30.11	29.97	29.74
Mean ± σ	97.55 ± 1.89	31.80	29.74	28.89	28.47	28.22 ± 1.70

Table 4 – Standard GA final run results per generation snapshot. ★ = overall best

The convergence profile is consistent with the expected behaviour of a low-mutation GA: rapid RMSE reduction in early generations transitions to progressively slower gains beyond generation 1 000 (fine-tuning phase). Run 07 is an outlier converging to 32.26; the remaining nine runs cluster tightly between 25.96 and 29.74, demonstrating the robustness of the chosen configuration.

8. Conclusion

A real-valued GA successfully produces a recognisable triangle-based approximation of Vermeer's Girl with a Pearl Earring. With 100 opaque triangles and 5 000 generations, the dominant structural features of the original; face profile, headscarf, dark background, pearl earring, illuminated collar, are clearly visible in the output (Figure 3).

A systematic ablation study across 300 runs identified mutation type as the dominant operator: Gaussian perturbation ($\sigma = 15$, $\text{mut_prob} = 0.01$) is essential for incremental refinement and decisively outperforms random-reset across all other operator settings. Tournament selection ($k = 5$) provides appropriate selection pressure without the stagnation risk of proportionate methods. Crossover type has minimal impact on this problem. Coverage-based initialisation was empirically ruled out; the alpha-channel variant shows directional promise but was not extended to the full run budget.

The final Standard GA configuration (tournament $k = 5$, uniform crossover, Gaussian mutation, $x_0 = 0.9$, $\text{mut} = 0.01$, $\text{pop} = 100$, 5 000 generations) achieved a mean RMSE of 28.22 and a best result of 25.96, a 71% reduction from the mean initial RMSE of 97.55. The Island Model variant shows lower variance across runs (std 1.20 vs 1.70) and a better worst case (31.63 vs 32.26), suggesting improved robustness; however, with only 10 runs per configuration this difference should be interpreted as directional rather than statistically confirmed.